

K13-A, NetherlandsWMO: **062520**Lat: **53.218N**Lon: **3.220E**Elev: **98**StdP: **14.64**Time Zone: **1.00 (EUC)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.8	34.1	21.2	16.0	34.4	24.2	18.5	35.9	45.3	44.4	42.3	44.8	19.8	90	0.836

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	68.1	64.5	66.5	63.4	65.6	62.6	65.2	67.3	64.0	66.1	63.2	65.3	15.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	90.1	66.6	62.9	86.2	65.6	62.1	83.7	65.0	30.1	67.6	29.3	66.4	28.7	65.2	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
40.9	36.9	33.5	DB	30.9	70.5	3.9	2.0	28.1	71.9	25.9	73.1	23.7	74.2	20.9	75.6
			WB	27.7	66.2	4.2	1.6	24.7	67.3	22.3	68.3	19.9	69.2	16.9	70.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	43.2	41.9	43.1	46.4	51.2	56.1	60.7	62.9	60.9	56.1	50.4	45.9
	DBStd	8.02	4.11	4.10	3.54	3.23	3.08	2.81	2.68	2.41	2.82	3.73	3.77	4.14
	HDD50	964	211	225	213	114	23	0	0	0	0	4	38	136
	HDD65	4894	675	645	678	557	428	268	137	75	125	276	437	593
	CDD50	1561	0	0	0	7	60	183	332	399	328	193	51	8
	CDD65	14	0	0	0	0	0	0	3	8	3	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.7	20.9	19.4	18.5	16.4	16.6	14.9	14.7	14.7	16.3	19.4	19.6	20.8
Precipitation	PrecAvg	38.4	3.7	2.8	2.6	1.8	2.0	2.5	2.9	3.7	3.7	4.1	4.3	4.3
	PrecMax	49.7	6.5	5.7	5.5	3.9	3.5	5.3	6.8	8.2	7.6	7.5	7.6	7.2
	PrecMin	28.5	0.5	0.9	0.4	0.1	0.9	0.7	0.5	0.8	0.7	1.5	0.7	2.0
	PrecStd	5.2	1.5	1.2	1.2	0.9	0.7	1.2	1.4	1.8	1.7	1.7	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	50.6	51.1	55.5	59.7	65.7	69.7	70.5	69.4	64.8	60.4	55.5
		MCWB	49.7	47.5	48.0	50.9	55.5	61.1	64.7	65.5	65.3	61.2	57.4	52.7
	2%	DB	51.0	49.2	49.4	52.9	57.5	62.3	66.9	68.5	66.6	63.0	58.8	53.9
		MCWB	48.7	46.7	47.1	49.8	54.3	59.3	63.7	64.9	63.1	60.0	55.9	51.3
	5%	DB	49.7	48.2	48.3	51.6	56.3	60.7	65.7	67.1	65.9	62.1	57.0	52.9
		MCWB	47.5	46.1	46.2	49.0	53.4	58.2	62.8	64.0	62.5	59.1	53.9	50.2
	10%	DB	48.3	47.1	47.1	50.4	55.1	59.4	64.4	66.0	64.7	60.9	55.5	51.3
		MCWB	46.1	45.0	45.1	48.2	52.6	57.0	61.7	63.1	61.4	58.0	52.2	48.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	48.5	48.8	51.8	56.7	61.9	65.7	67.0	65.7	61.9	58.4	53.6
		MCDB	51.8	49.5	50.0	53.9	58.4	64.5	68.3	68.5	68.0	63.8	60.2	54.8
	2%	WB	49.1	47.3	47.6	50.3	55.1	59.9	64.3	65.6	63.9	60.7	56.2	52.2
		MCDB	50.5	48.6	48.9	52.1	56.8	61.6	66.5	67.6	66.0	62.8	58.3	53.9
	5%	WB	47.8	46.3	46.5	49.4	54.0	58.7	63.2	64.6	63.3	59.7	54.5	50.6
		MCDB	49.4	47.8	47.8	51.1	55.8	60.3	65.3	66.7	65.4	61.8	56.8	52.5
	10%	WB	46.4	45.2	45.5	48.4	53.0	57.5	62.0	63.6	62.2	58.6	52.8	49.1
		MCDB	48.0	46.8	46.8	50.1	54.9	59.2	64.0	65.8	64.5	60.8	55.2	51.1
Mean Daily Temperature Range	5% DB	MDBR	3.9	3.7	3.6	3.5	3.3	3.3	3.3	3.2	3.5	3.9	4.0	4.1
		MCDBR	4.8	4.1	4.2	4.9	4.8	4.9	4.8	4.2	3.9	4.5	4.8	4.8
		MCWBR	5.8	4.1	4.0	3.4	3.7	4.1	3.5	3.7	4.4	5.3	5.8	5.7
	5% WB	MCDBR	4.6	4.0	4.0	4.4	4.6	4.9	4.6	3.8	3.7	4.4	4.6	4.5
		MCWBR	5.7	4.3	4.0	3.4	3.9	4.3	3.5	3.8	4.4	5.4	5.8	5.8
Clear-Sky Solar Irradiance	taub	0.332	0.348	0.377	0.402	0.400	0.403	0.408	0.398	0.385	0.363	0.338	0.322	
	taud	2.326	2.364	2.312	2.281	2.362	2.366	2.361	2.401	2.429	2.424	2.366	2.314	
	Ebn at Noon	200	238	254	262	270	269	266	262	250	229	197	180	
	Edh at Noon	19	25	32	37	36	36	36	33	29	23	18	16	
All-Sky Solar Radiation	RadAvg	204	434	857	1414	1762	1842	1778	1495	1033	554	247	153	
	RadStd	19	72	73	134	113	99	184	104	77	43	22	16	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page